Digital Design 101 | 2026-1

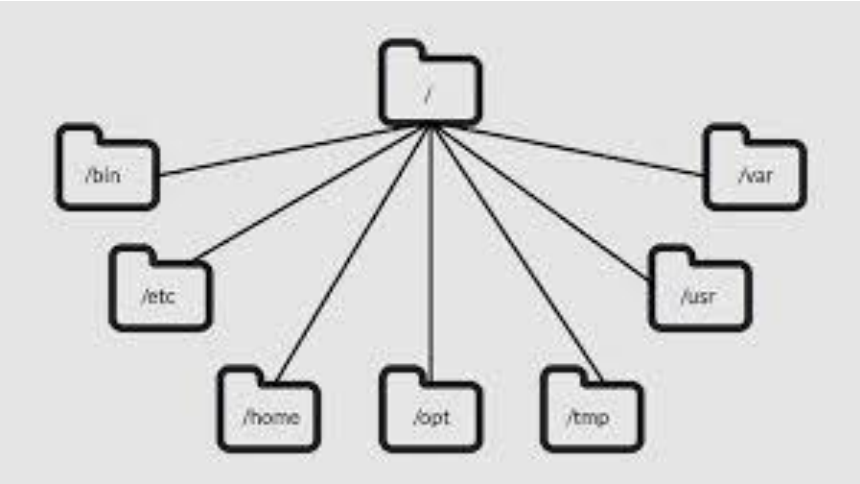
Session2 : Basic Linux Command

1. Basic Linux Operation

Directory

```
cd  {directory}
pwd
```

- Use Absolute PATH
- Use Relative PATH
- Use user’s home



Manual

man {command}

- Search Keyword
- Moving Back & Forth
- Quit Manual

```
LSOF(8)                    System Manager's Manual                    LSOF(8)

NAME
       lsof - list open files

SYNOPSIS
       lsof  [  -?abChlnNOPRtUvVX ] [ -A A ] [ -c c ] [ +c c ] [ +|-d d ] [
       +|-D D ] [ +|-e s ] [ +|-f [cfgGn] ] [ -F [fl ] [ -g [s] ] [ -i  [i]
       ] [ -k k ] [ -K k ] [ +|-L [l] ] [ +|-m m ] [ +|-M ] [ -o [o] ] [ -p
       s ] [ +|-r [t[m<fmt>]] ] [ -s [p:s] ] [ -S [t] ] [ -T [t] ] [ -u s ]
       [ +|-w ] [ -x [fl] ] [ -z [z] ] [ -Z [Z] ] [ -- ] [names]

DESCRIPTION
       Lsof  revision  4.87  lists  on its standard output file information
       about files opened by processes for the following UNIX dialects:

            Apple Darwin 9 and Mac OS X 10.[567]
            FreeBSD 4.9 and 6.4 for x86-based systems
            FreeBSD 8.2, 9.0 and 10.0 for AMD64-based systems
            Linux 2.1.72 and above for x86-based systems
            Solaris 9, 10 and 11

       (See the DISTRIBUTION section of this manual page for information on
       how to obtain the latest lsof revision.)

       An  open  file  may  be a regular file, a directory, a block special
       file, a character special  file,  an  executing  text  reference,  a
       library,  a  stream  or a network file (Internet socket, NFS file or
       UNIX domain socket.)  A specific file or all the  files  in  a  file
       system may be selected by path.

       Instead of a formatted display, lsof will produce output that can be
       parsed by other programs.  See the -F, option description,  and  the
       OUTPUT FOR OTHER PROGRAMS section for more information.

       In  addition  to  producing  a  single output list, lsof will run in
       repeat mode.  In repeat mode it will  produce  output,  delay,  then
       repeat  the output operation until stopped with an interrupt or quit
       signal.  See the +|-r [t[m<fmt>]] option description for more infor-
       mation.

OPTIONS
       In  the  absence of any options, lsof lists all open files belonging
       to all active processes.
 Manual page lsof(8) line 1 (press h for help or q to quit)
```

Clear the screen

```
clear
```

Listing Directory

ls –option {directory}

- Listing Directory
- Read Output
- Options
  - Hidden File / Directory
  - Verbose

```
dr-xr-xr-x.  17 root root   4096 Aug 13 10:31 .
dr-xr-xr-x.  17 root root   4096 Aug 13 10:31 ..
-rw-r--r--.   1 root root 439505 Aug  9 16:36 .readahead
lrwxrwxrwx.   1 root root      7 Aug  9 16:20 bin -> usr/bin
dr-xr-xr-x.   4 root root   4096 Aug 10 13:32 boot
drwxr-xr-x.  21 root root   3560 Aug  9 16:34 dev
drwxr-xr-x. 148 root root   8192 Aug 17 02:01 etc
drwxr-xr-x.   5 root root     66 Aug 16 00:12 home
lrwxrwxrwx.   1 root root      7 Aug  9 16:20 lib -> usr/lib
lrwxrwxrwx.   1 root root      9 Aug  9 16:20 lib64 -> usr/lib64
drwxr-xr-x.   2 root root      6 Dec 15  2017 media
drwxr-xr-x.   2 root root      6 Dec 15  2017 mnt
drwxr-xr-x.   4 root root     41 Aug  9 16:56 opt
dr-xr-xr-x. 866 root root      0 Aug  9 16:34 proc
dr-xr-x---.  18 root root   4096 Aug 17 02:01 root
drwxr-xr-x.  45 root root   1360 Aug 18 06:54 run
lrwxrwxrwx.   1 root root      8 Aug  9 16:20 sbin -> usr/sbin
drwxr-xr-x.   2 root root      6 Dec 15  2017 srv
dr-xr-xr-x.  13 root root      0 Aug  9 16:34 sys
drwxrwxrwt.  29 root root  20480 Aug 18 16:01 tmp
drwxr-xr-x.  13 root root    207 Aug  9 16:20 usr
drwxr-xr-x.  20 root root   4096 Aug  9 16:34 var
```

Init File[s] / Directorie[s]

```
touch {file_name}
```

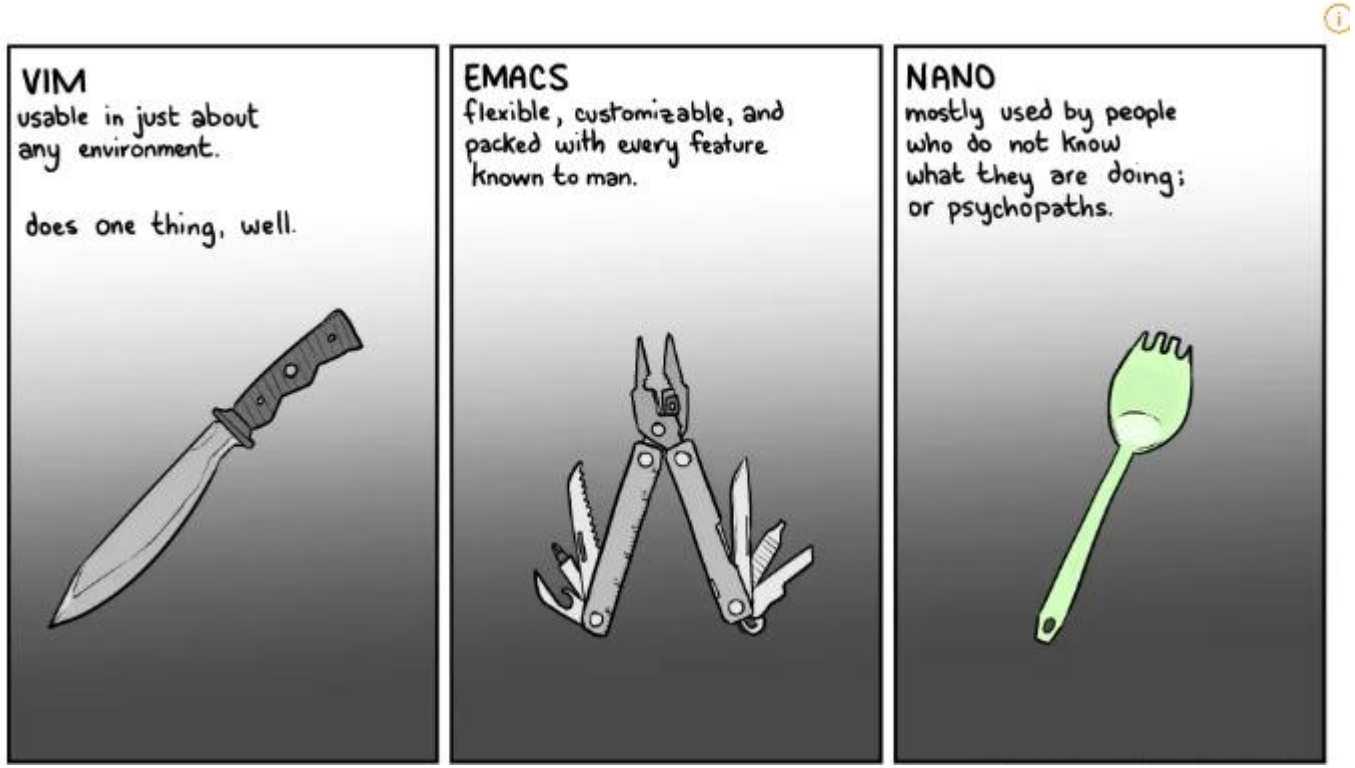
- Make multiple files

```
mkdir -option {directory_name}
```

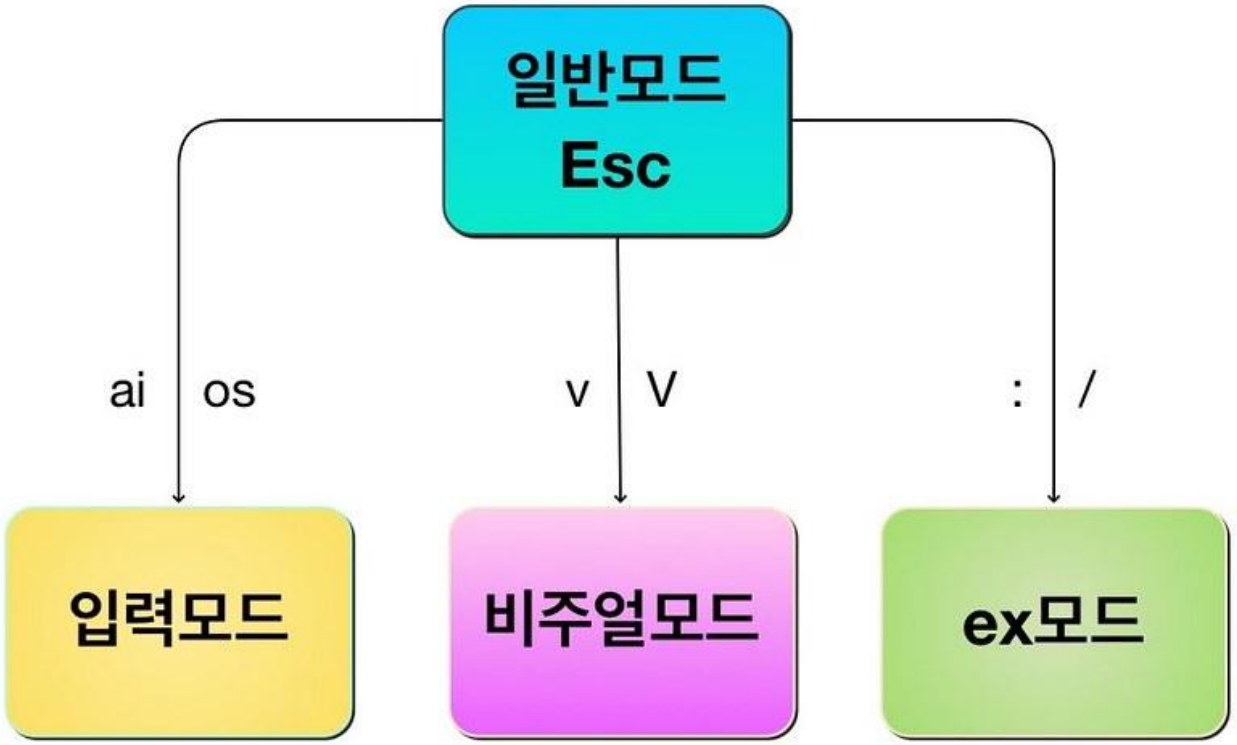
- Nested directory

VI(M)

Unix/Linux 자체에 내장된 문서 편집기 프로그램.
강력한 문서 편집과 Customization을 통한 사용확장성을 보여준다.



VI(M): Operation Mode



VI(M): insert mode

입력모드는 텍스트를 입력(편집)하는 모드이다

- a : 커서위치 다음 칸에 입력
- i : 커서의 위치에 입력
- o : 커서의 다음 행에 입력
- s : 커서 위치의 한글자를 삭제 후 입력

VI(M): visual mode

비쥬얼 모드는 블록 단위를 선택하여 작업하는 모드이다.

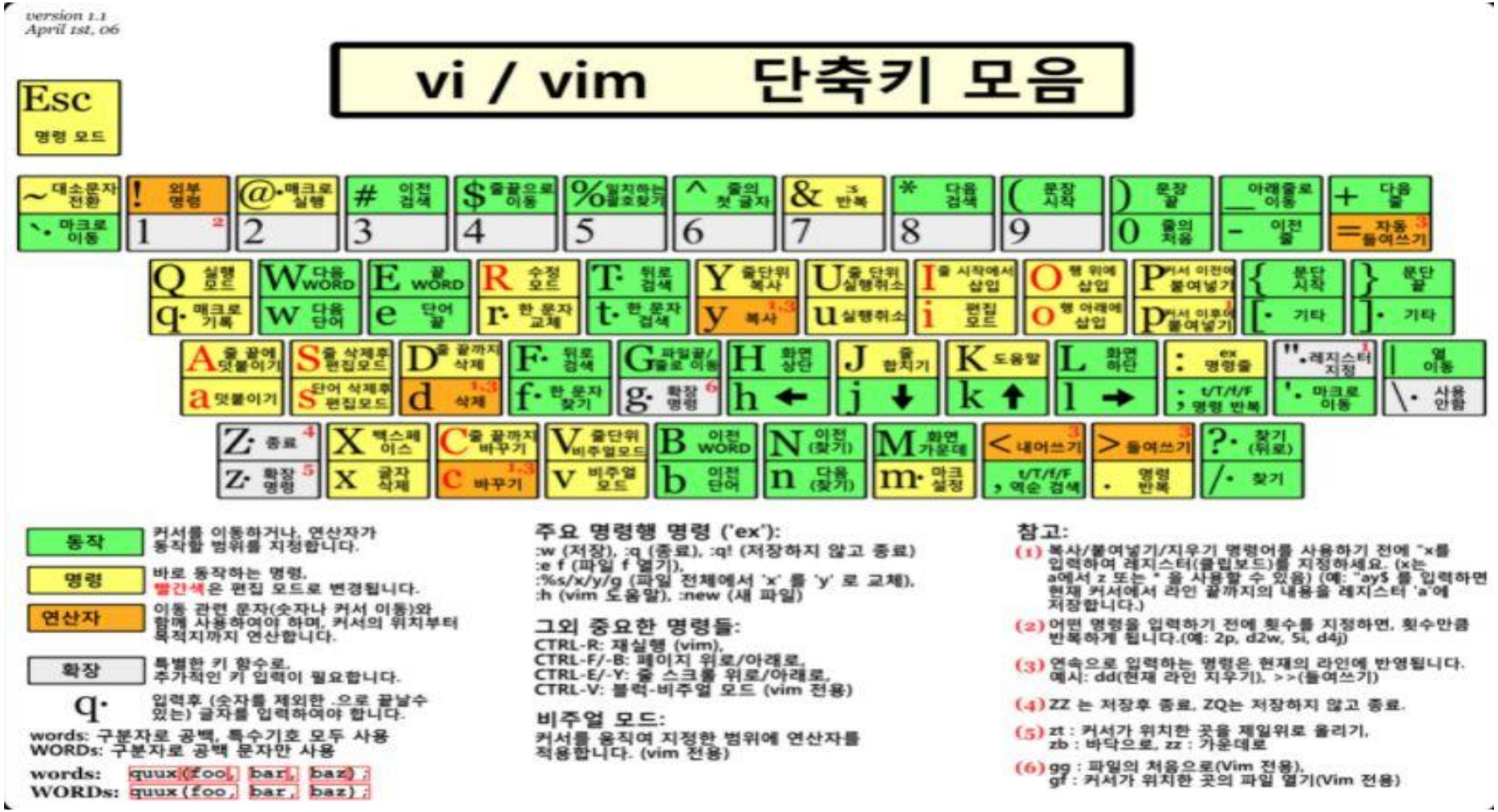
- v : Normal Visual Mode
- V : Visual Line Mode
- d: 선택된 내용 삭제 -> Normal Mode
- c: 선택된 내용 삭제 -> Insert Mode
- y: 선택된 내용의 복사 -> Normal Mode
- J: 선택된 모든 라인을 순서대로 결합 -> Normal Mode
- ~ : 선택된 내용의 case를 변경 -> Normal Mode
- u : 선택된 내용을 소문자로 변경 -> Normal Mode
- U : 선택된 내용을 대문자로 변경 -> Normal Mode

VI(M): ex mode

ex 모드는 작업을 명령하는 모드이다

- :w - 저장
- :w file.txt – file.txt로 저장
- :w >> file.txt – file.txt에 append하여 저장
- :q – VIM 종료
- :q! – 강제 종료
- :wq! – 저장하고 강제 종료
- :e file.txt – file.txt 를 불러옴

VI(M) 단축키



- [VIM 연습]

Copy and Paste

```
cp –option {src} {dst}
rm –option {src} {dst}
```

- Directory
- Recursive Operation

Duplicity of “mv” command

mv [src] [dst]
mv [src1] [src2]

- Moving directory
- Change directory’s name

```
dr-xr-xr-x.  17 root root   4096 Aug 13 10:31 .
dr-xr-xr-x.  17 root root   4096 Aug 13 10:31 ..
-rw-r--r--.   1 root root 439505 Aug  9 16:36 .readahead
lrwxrwxrwx.   1 root root      7 Aug  9 16:20 bin -> usr/bin
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lrwxrwxrwx.   1 root root      9 Aug  9 16:20 lib64 -> usr/lib64
drwxr-xr-x.   2 root root      6 Dec 15  2017 media
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drwxr-xr-x.  13 root root    207 Aug  9 16:20 usr
drwxr-xr-x.  20 root root   4096 Aug  9 16:34 var
```

Printing the contents in the file

```
cat {file}
more / less {file}
```

- Quitting printed page
- Move back and forth through pages

```
head {file}
tail {file}
```

2. Intermediate Linux Operation

Shell(1)

Shell은 리눅스 커널과 사용자를 연결해 주는 인터페이스
명령어(프로그래밍 언어)이다

echo $0
: 현재 사용 중인 사용자의 Shell을 출력한다

- Bash Shell
- C-Shell for EDA

Shell(2)

source {shell_file}
: Shell 파일의 내용을 적용한다

- Shell 적용의 우선순위

Alias

```
alias {shorten_customized_cmd} = “original_long_cmd”
```

- 단축 명령어를 설정하는 방법
- 즐겨찾기

Link

- Hard Link: ln [src] [dst]
- Soft Link: ln –s [src] [dst]

- 복사와 링크의 다른점
- 원본 파일 변경/삭제
- Hard Link보다 Soft(Symbolic) Link가 선호되는 이유

grep

- grep –option {serach_pattern} {file or path}

Compress using tar

- tar –xzvf {compressed_file}
- tar –czvf {compressed file} {src_directory}

- Process of Compressing
- Extension name of compressed file

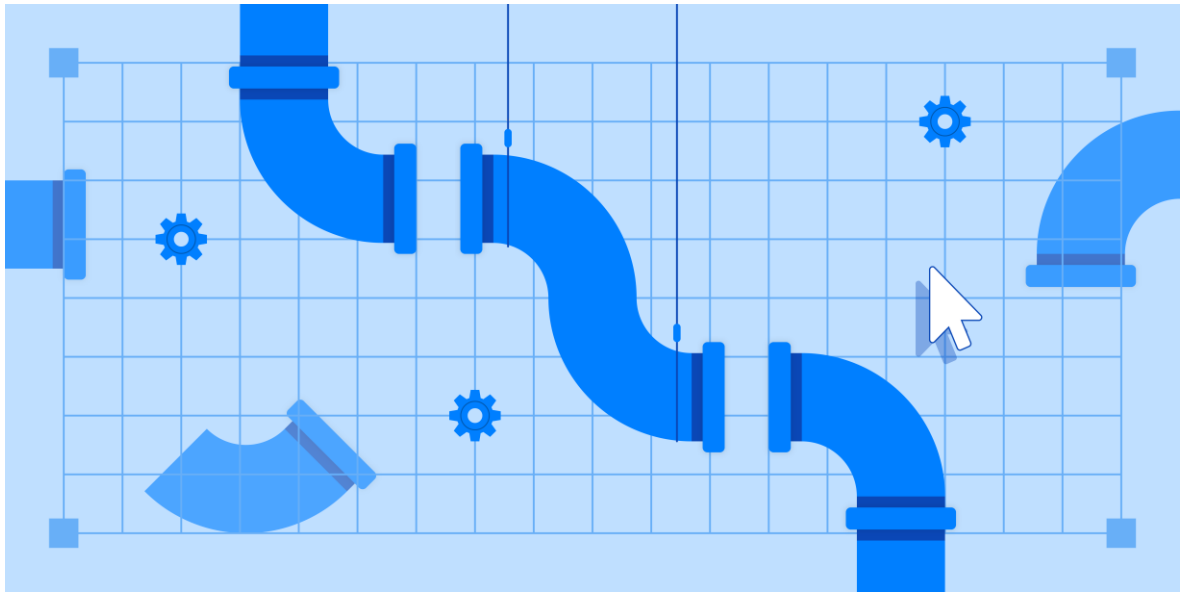
Logical Concatenation of commands

- command1 ; command2
- command1 && command2
- command1 || command2

Command Pipelining

```
command1 | command2
```

- Pipeline 개념의 공학적 응용



Redirection IO

```
command > file
command >> file
```

- Standard Input / Output

Control Process

ps : prints user’s whole process
kill : terminate user’s PID

- Process란?
- PID(Process ID)

Permission

rwx: read, write, execution
ugo: user, group, other[s]

chmod –option permission_masking_bits {file or directory}

- Octal 사용법
- +/- rwx 사용법
- Recursive Operation